

A Multimodal Contrastive and Transfer Learning-Based Image Restoration Model for Multiple Adverse Weather Driving Scenes

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Abstract—Adverse weather conditions like rain, fog, and snow significantly hinder perception in autonomous driving systems. This paper proposes a multimodal contrastive learning and transfer learning-based adaptive image restoration method for multiple adverse weather conditions. By integrating image and textual information, our method enhances robustness to diverse weather scenarios. Specifically, we first fine-tune a contrastive language-image pre-trained model to develop a multimodal image classifier capable of recognizing adverse weather conditions. Subsequently, an encoder-decoder-based restoration network is employed, where cross-attention layers incorporate textual conditional information, enabling the network to perceive weather variations. An adaptive restoration strategy is then applied to target specific noise characteristics associated with different weather conditions. Experiments on Rain Cityscapes, Foggy Cityscapes, and Snow Cityscapes show our model outperforms task-specific and All-in-One methods in visual and real-time performance, providing an efficient and robust solution for autonomous driving in complex environments.

Index Terms—Image restoration, image deraining, image dehazing, image desnowing, multimodal contrastive learning.

I. INTRODUCTION

WITH the rapid development of artificial intelligence, autonomous driving technology has increasingly become a significant research focus within transportation [1]. Utilizing advanced sensors, algorithms, and communication technologies, autonomous vehicles can navigate complex environments, enhancing road safety and optimizing traffic flow, thereby providing robust technical support for smart city development. However, adverse weather conditions remain a critical challenge for autonomous driving systems [2], [3], [4]. Rain, fog, and snow degrade visual sensors, reduce image clarity, and increase the likelihood of detection errors, thereby compromising driving safety [5], [6], [7]. Developing effective image restoration techniques is therefore critical for enhancing the robustness of autonomous driving under complex weather conditions.

Research on image restoration has initially focused on single-degradation scenarios. For deraining, attention-based generative adversarial networks (GAN) [8], lightweight recursive networks [9], and more recent efficient designs such as ESTINet [10]

have shown effectiveness. Fog removal has benefited from approaches like AOD-Net [11] and FFA-Net [12], while snow removal has been advanced through multi-stage frameworks such as DesnowNet [13]. Beyond weather-related degradations, deblurring networks such as DBLRNet [14] and MC-Blur [15] have further demonstrated the progress of restoration backbones on motion blur. However, despite these advances, single-task models remain limited when facing diverse and unpredictable conditions, motivating the development of unified approaches.

To address this issue, All-in-One image restoration methods have emerged, aiming to handle multiple degradations within a unified model. Multi-branch networks [16], teacher-student frameworks [17], and transformer-based variants such as MB-TaylorFormer V2 [18] and GridFormer [19] have demonstrated promising results. However, constructing a shared representation that generalizes across heterogeneous degradations remains challenging. In addition, many of these methods rely on explicit weather priors or require high computational cost, limiting their deployment in real-time driving scenarios. Recent studies further explored the use of pre-trained multimodal models to guide restoration. Xu et al. [20] leveraged priors from large-scale models to boost general image restoration, while DA-CLIP [21] adapted contrastive language-image pretraining (CLIP) for universal restoration. These works highlight the potential of semantic priors, though they are not specifically designed for multi-weather traffic scenes. Meanwhile, hybrid losses have long been applied in image restoration [22], [23], but their adaptation to multi-weather tasks has rarely been investigated.

This paper presents a unified model for restoring traffic scene images under multiple adverse weather conditions. The approach integrates multimodal contrastive learning with transfer learning to recognize weather types and adapt restoration accordingly. By introducing textual semantic cues through a cross-attention mechanism, the model leverages both visual and language representations to guide degradation-aware enhancement. This design enhances the reliability and stability of environmental perception in autonomous driving. The main contributions are as follows:

- 1) A unified image restoration model tailored to multi-weather traffic scenes, improving robustness across rain, fog, and snow.
- 2) A fine-tuned CLIP-based multimodal classifier is incorporated with cross-attention to provide weather-aware semantic guidance for adaptive restoration.
- 3) Experiments demonstrate competitive restoration accuracy with real-time efficiency, reaching 57.03, 62.41, and 62.44 frames per second on Rain, Foggy, and Snow Cityscapes.

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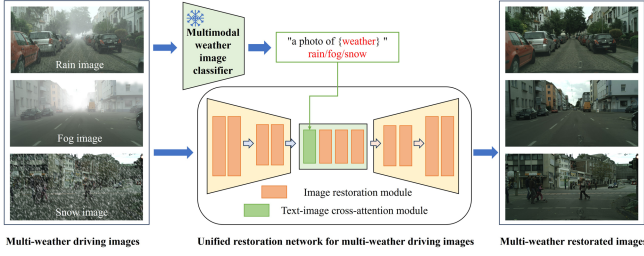


Fig. 1. Framework of the multi-weather image restoration model based on multimodal contrastive learning.

II. METHODOLOGY

A. Framework of the Proposed Model

In this paper, we propose a unified adverse weather restoration model that integrates multimodal contrastive learning with transfer learning. As shown in Fig. 1, the CLIP model is used to extract joint visual-textual features, followed by transfer learning to build a weather-specific image classifier. Integrating this classifier with the restoration network enables accurate recognition and adaptive restoration of rain, fog, and snow conditions.

B. Multimodal Contrastive Learning and Transfer Learning-Based Image Classifier

Multimodal contrastive models trained on large-scale image-text data demonstrate strong semantic understanding but struggle to distinguish complex weather types in autonomous driving scenarios [24]. To overcome this, we fine-tune a domain-specific multimodal classifier on weather-labeled data, enabling accurate recognition and guiding conditional restoration.

As shown in Fig. 2, the process involves: (1) Extracting features from images and weather text using frozen CLIP encoders; (2) Applying a lightweight classification head with cosine similarity for category prediction; (3) Fine-tuning the classifier to adapt to weather-specific noise while retaining pretrained knowledge. This approach significantly enhances classification performance, even under limited data conditions, improving weather recognition for robust image restoration.

C. Multimodal Image Classifier-Based Unified Multi-Weather Restoration Network

As shown in Fig. 3, we propose a unified multi-weather image restoration network guided by a multimodal classifier. The architecture follows a U-Net design and consists of four main components: a multimodal weather classifier, an encoder, a cross-modal attention module, and a decoder.

1) *Multimodal Weather Classifier*: This module uses a pre-trained classifier with frozen parameters to predict the weather condition and generate a corresponding textual description based on the input image.

2) *Encoder*: The input image is first embedded into patch tokens and passed through four stages. Stages 1 and 3 extract detailed features using depthwise separable convolutions, channel attention, feedforward networks, and residual connections. Stages 2 and 4 perform downsampling and capture higher-level semantic information.

3) *Cross-Modal Attention Module*: Positioned between the encoder and decoder, this module injects semantic text features

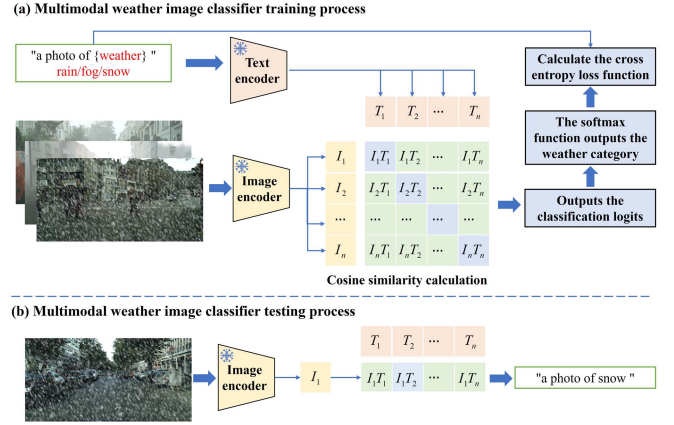


Fig. 2. The training and testing process of the multimodal weather image classifier.

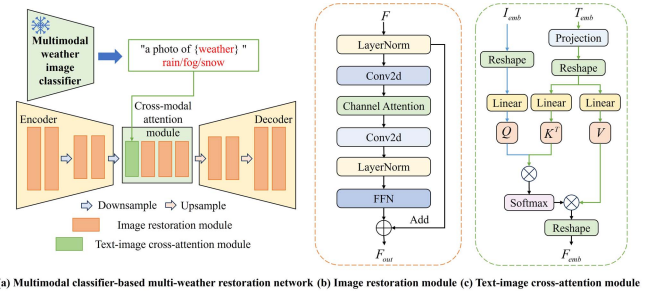


Fig. 3. Multimodal image classifier-based multi-weather restoration network.

into visual features. The encoder output I_{emb} is reshaped as a sequence Q , and the corresponding weather text feature is projected to K and V . The attention computation is formulated as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{C}}\right)V \quad (1)$$

The result is reshaped and fused with the original image feature via a residual connection to guide restoration.

4) *Decoder*: The decoder upsamples feature maps and fuses low-level details via skip connections. It refines features through image restoration modules and outputs the final image. A residual connection with the input ensures better detail and structure preservation.

D. Loss Function

The proposed model uses a joint loss function to balance pixel-level accuracy and structural consistency. It combines L1 loss and structural similarity index measure (SSIM) loss with equal weight, defined as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{L1}} + \mathcal{L}_{\text{SSIM}} \quad (2)$$

This simple formulation ensures that the model simultaneously focuses on preserving fine details and overall structural similarity during training.

TABLE I
QUANTITATIVE RESULTS ON THE RAIN CITYSCAPES DATASET

Method	PSNR↑	SSIM↑	Time(s)↓	FPS↑	MPx/s↑
AttnGAN	16.32	0.8374	2.189	0.4569	0.9582
DuRN	25.58	0.9604	2.771	0.3609	0.7569
PReNet	28.77	0.9697	0.9294	1.076	2.257
IDT	33.30	0.9855	7.474	0.1338	0.2806
AirNet	18.24	0.8524	8.340	0.1199	0.2514
TKL	34.19	0.9857	2.039	0.4904	1.028
Ours	35.31	0.9882	0.0175	57.03	119.6

III. EXPERIMENTS

A. Dataset and Experiment Settings

This study uses three datasets derived from Cityscapes [25]: Rain Cityscapes [26], Foggy Cityscapes [27], and Snow Cityscapes [28], which represent rain, fog, and snow conditions. In total, the combined dataset includes 24357 training and 7763 testing image pairs. Rain and Foggy Cityscapes have a resolution of 2048×1024 , while Snow Cityscapes is 512×256 . To ensure comparability, all images are cropped into 256×256 patches for training, while evaluation is performed at the original resolution of each dataset. All experiments are implemented in PyTorch on an NVIDIA RTX A6000 GPU with a batch size of 16, trained for 200 epochs using the AdamW optimizer and a learning rate of $1e-4$.

B. Evaluation Metrics

To assess restoration performance under adverse weather, we use two standard image quality metrics: peak signal-to-noise ratio (PSNR) and SSIM. Higher values indicate better results. Runtime efficiency is also evaluated using average inference time, frames per second (FPS), and million pixels per second (MPx/s), providing a clear measure of real-time applicability in autonomous driving.

C. Comparative Experiments and Analysis

To validate the effectiveness of the proposed multimodal contrastive and transfer learning-based adaptive restoration model, we conduct comparative experiments on rain, fog, and snow scenarios. Both task-specific and All-in-One models are included for comparison.

1) *Results on the Rain Cityscapes Dataset:* For the rain removal task, we include several representative task-specific models, such as GAN-based (AttnGAN [8]), CNN-based (DuRN [29], PReNet [9]), and Transformer-based (IDT [30]) approaches. Additionally, we evaluate two All-in-One general-purpose image restoration models (AirNet [16] and TKL [17]). On the Rain Cityscapes dataset, as shown in Table I, our model achieves the highest performance with 35.31 dB PSNR and 0.9882 SSIM, surpassing all baselines. Notably, it outperforms IDT by 2.01 dB in PSNR and 0.0027 in SSIM. Among All-in-One models, TKL is the best performer with 34.19 dB PSNR and 0.9857 SSIM, still lagging behind our results.

In terms of efficiency, our method significantly outperforms others with an average inference time of 0.0175 seconds per image, far faster than IDT, AirNet, and even the fastest model PReNet. It reaches 57.03 FPS and 119.6 MPx/s, highlighting its strong real-time capabilities and suitability for high-resolution autonomous driving scenarios.

Fig. 4 shows visual comparisons of rain removal. Our model effectively removes rain streaks while preserving details and

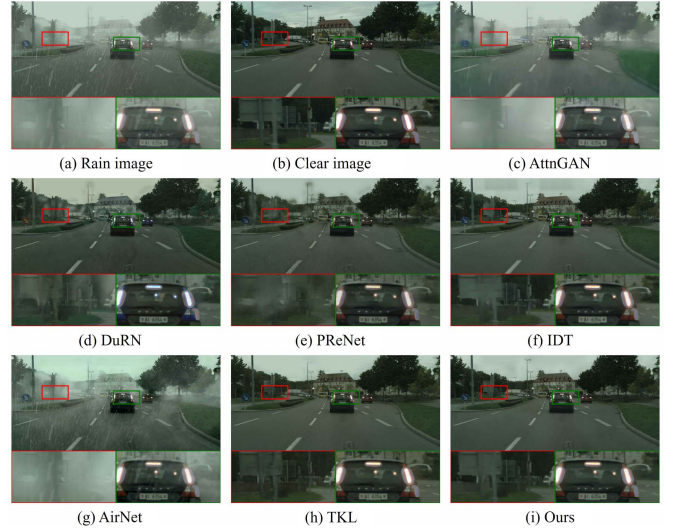


Fig. 4. The rain removal effect on the rain cityscapes dataset.

TABLE II
QUANTITATIVE RESULTS ON THE FOGGY CITYSCAPES DATASET

Method	PSNR↑	SSIM↑	Time (s)↓	FPS↑	MPx/s↑
DCP	16.93	0.8219	30.93	0.032	0.067
AOD-Net	19.58	0.8971	0.0013	756.6	1586
FFA-Net	29.91	0.9747	0.5377	1.860	3.900
Dehazer	26.49	0.9606	4.742	0.211	0.442
AirNet	17.90	0.8231	8.3612	0.1196	0.2508
TKL	33.09	0.9859	1.969	0.5079	1.065
Ours	34.26	0.9896	0.0160	62.41	130.9

color fidelity. In contrast, AttnGAN and AirNet leave streaks and blur, DuRN and PReNet retain fog or distort colors, and IDT introduces mosaic artifacts. Overall, our results are cleaner and more realistic, closely matching the ground truth.

2) *Results on the Foggy Cityscapes Dataset:* To assess performance on the dehazing task, we conduct comparisons on the Foggy Cityscapes dataset using both traditional and state-of-the-art models, including DCP [31], AOD-Net [11], FFA-Net [12], Dehazer [32], as well as All-in-One methods like AirNet [16] and TKL [17]. As shown in Table II, our model achieves the best results with 34.26 dB PSNR and 0.9896 SSIM, outperforming all baselines. TKL comes closest but still trails by 1.17 dB in PSNR.

In terms of efficiency, although AOD-Net stands out with the fastest inference speed, its dehazing quality is inadequate, limiting its practical use. In contrast, our model achieves an excellent balance, delivering 62.41 FPS and 130.9 MPx/s while maintaining leading image quality, making it well-suited for real-time autonomous driving applications.

As illustrated in Fig. 5, traditional methods like DCP produce overexposed results, and AOD-Net and AirNet leave fog residues. FFA-Net and Dehazer perform better but lack texture sharpness. In contrast, our model effectively removes fog while restoring clear structures and realistic textures, offering the best visual quality for autonomous driving scenarios.

3) *Results on the Snow Cityscapes Dataset:* To evaluate performance under snowy conditions, we test the proposed model on the Snow Cityscapes dataset and compare it with both task-specific snow removal methods DesnowNet [13], DesnowGAN [33], HDCWNet [34], DDMSNet [28] and All-in-One models. As shown in Table III, our model achieves the best results with 39.16 dB PSNR and 0.9870 SSIM, outperforming all baselines.

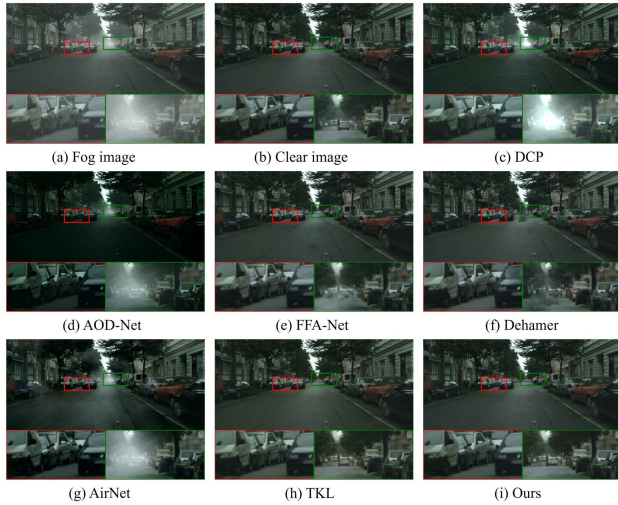


Fig. 5. The fog removal effect on the foggy cityscapes dataset.

TABLE III
QUANTITATIVE RESULTS ON THE SNOW CITYSCAPES DATASET

Method	PSNR $\uparrow$	SSIM $\uparrow$	Time (s) $\downarrow$	FPS $\uparrow$	MPx/s $\uparrow$
DesnowNet	27.78	0.9405	0.7605	1.315	0.1724
DesnowGAN	30.32	0.9414	0.0140	71.50	9.372
RESCAN	38.32	0.9809	0.0085	117.9	15.46
DDMSNet	38.56	0.9843	0.2163	4.623	0.6059
AirNet	22.40	0.8747	0.4310	2.320	0.3041
TKL	37.37	0.9830	0.1189	8.410	1.102
Ours	39.16	0.9870	0.0160	62.44	8.184

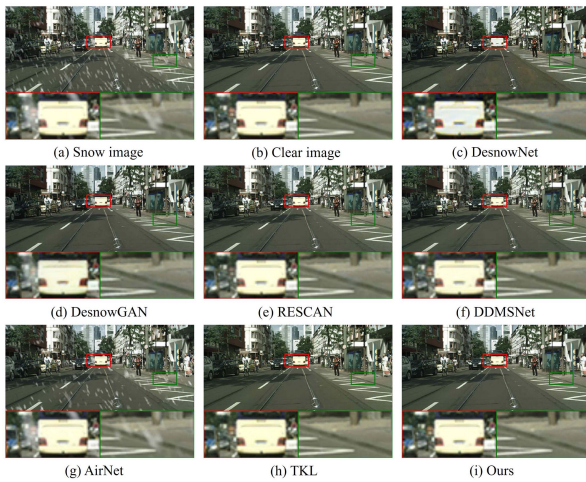


Fig. 6. The snow removal effect on the snow cityscapes dataset.

While DDMSNet offers good quality, its slower runtime limits practicality. Our model runs at 0.016 s per image, reaching 62.44 FPS, combining high accuracy with real-time performance.

Visual results in Fig. 6 show that models like AirNet and DesnowGAN leave residual snow or blurred regions, while DDMSNet suffers from over-smoothing. In contrast, our model removes snow more effectively while preserving details and edges, producing cleaner and more realistic images.

Overall, the proposed model delivers state-of-the-art performance and real-time efficiency across all three weather conditions. By combining multimodal contrastive learning with

TABLE IV
ABLATION STUDY ON CORE MODULES OF THE PROPOSED MODEL

Method	Rain		Fog		Snow	
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
Baseline	32.94	0.9782	32.19	0.9808	35.61	0.9660
+ Multimodal classifier	34.57	0.9840	33.69	0.9859	37.31	0.9786
+ Classifier + Joint loss	35.31	0.9882	34.26	0.9896	39.16	0.9870

TABLE V
ABLATION STUDY ON CLASSIFIER AND FUSION DESIGN CHOICES

Method	Rain		Fog		Snow	
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
Image-only classifier	34.53	0.9875	33.53	0.9891	38.84	0.9868
Feature concatenation	34.17	0.9866	33.07	0.9879	38.53	0.9860
Ours	35.31	0.9882	34.26	0.9896	39.16	0.9870

transfer learning, it outperforms both task-specific and All-in-One models in flexibility, accuracy, and deployment speed, making it highly suitable for autonomous driving and similar real-world scenarios.

D. Ablation Study

To evaluate the contribution of each module in the proposed model, we first conduct a stepwise ablation using the baseline restoration network, the addition of the multimodal classifier, and the further integration of the joint loss. As shown in Table IV, introducing the classifier consistently improves performance across all three weather conditions. For example, the PSNR on rainy images increases by 1.63 dB, showing the benefit of incorporating cross-modal attention guided by weather-specific text features. Incorporating the joint loss further enhances both PSNR and SSIM, with the best result of 39.16 dB PSNR achieved on snowy scenes, confirming the importance of jointly preserving detail and structural fidelity.

To further validate the design choices, Table V compares different classifier and fusion strategies. Results show that using an image-only ResNet-50 classifier or replacing cross-attention with feature concatenation degrades restoration quality, highlighting the advantage of multimodal semantics and attention-based fusion. The multimodal classifier also provides stable semantic cues that improve restoration consistency across diverse weather types.

IV. CONCLUSION AND FUTURE WORK

This paper proposed an adaptive unified image restoration model for autonomous driving under adverse weather, integrating multimodal contrastive and transfer learning. A fine-tuned CLIP classifier provides weather-aware semantics through cross-attention with visual features, while a joint loss improves both detail reconstruction and structural fidelity. Experiments on Rain, Foggy, and Snow Cityscapes demonstrate superior performance over task-specific and unified methods, achieving 35.31 dB PSNR and 0.9882 SSIM in rain, with an average inference speed above 60 FPS across all weather conditions. While the model demonstrates strong robustness across diverse weather conditions, the issue of domain shift among datasets remains a potential limitation. Future work will explore domain adaptation and domain generalization methods to further enhance cross-dataset robustness. In addition, extending the framework to real-world mixed-weather scenarios and coupling it with downstream perception tasks such as detection and segmentation are promising directions for further research.

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